

Research progress and trend of aerobic exercise intervention in patients with knee osteoarthritis

A knowledge mapping study based on bibliometrics

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Abstract

Background: Knee osteoarthritis (KOA) is a prevalent chronic condition among middle aged and elderly individuals, characterized by joint pain, cartilage degeneration, and limited mobility, severely impacting quality of life. Aerobic exercise has been widely recognized as an effective non-pharmacological intervention to improve joint function and alleviate symptoms.

Methods: This bibliometric analysis explored global research trends related to aerobic exercise interventions for KOA from 2014 to 2024. A total of 273 articles were retrieved from the Web of Science Core Collection using predefined search strategies. Analytical tools including CiteSpace, SCImago Graphica, VOSviewer, and COOC14.9 were applied to examine publication outputs, keyword co-occurrence, and international collaboration networks.

Results: The United States, China, and Australia emerged as the leading contributors to research in this area, forming an active global collaborative network. High frequency research themes included “knee osteoarthritis,” “pain management,” and “quality of life,” while emerging hotspots were “Tai Chi,” “randomized controlled trial,” and “personalized intervention.” Over the past decade, research focus has shifted towards diversification of exercise modalities, individualized approaches, and long-term management strategies.

Conclusion: This study provides a comprehensive overview of global research developments in aerobic exercise interventions for KOA. The findings reveal evolving research trends and suggest future directions for academic exploration, offering valuable insights for researchers and healthcare professionals engaged in exercise-based KOA management.

Abbreviations: KOA = knee osteoarthritis, RCT = randomized controlled trial, WoS = Web of Science.

Keywords: aerobic exercise, bibliometrics, knee osteoarthritis, knowledge graph, research trends

1. Introduction

Knee osteoarthritis (KOA) is a chronic degenerative disease, which is mainly characterized by articular cartilage degeneration, bone hyperplasia and synovial inflammation, often leading to joint pain, limited movement and other symptoms.^[1] With the aggravation of global population aging, the incidence and prevalence of KOA have increased significantly, which has a serious impact on the quality of life and physical function of middle-aged and elderly people.^[2–4] Traditional treatments for KOA include drugs, physical therapy, and surgery, but these methods have limited effects and side effects,^[5] which cannot meet the long-term management needs of patients.^[5–8] Therefore, exploring safe and sustainable non-pharmaceutical interventions has become the focus of KOA prevention and treatment.

In recent years, aerobic exercise has attracted attention because of its effects in enhancing physical fitness, improving joint stability, relieving pain, and has gradually become one of the effective intervention methods for KOA.^[9,10] The common aerobic exercise methods include walking, water cycling, Tai chi, Wuqinxi Baduanjin, swimming, etc. Studies have shown that these exercises have certain effects in reducing the symptoms of KOA and improving the quality of life of patients.^[11–16] However, there is no consensus on the best intervention method, exercise intensity, and frequency of aerobic exercise for KOA patients, and there is obvious heterogeneity in the results of related studies. Therefore, it is of great practical significance to systematically sort out and analyze the existing research results of aerobic exercise intervention for optimizing the exercise prescription of KOA.

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Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

This study did not include any patient information. Thus, the requirement for ethics approval was waived.

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At present, more and more scholars have begun to pay attention to the application of aerobic exercise in patients with KOA. Studies have shown that low to moderate intensity aerobic exercise can effectively reduce joint pain, improve joint function, and reduce drug dependence in patients with KOA. However, studies vary greatly in exercise forms, intervention intensity, and efficacy evaluation indicators, resulting in inconsistent research results. In addition, the influence of individual differences (such as gender, age, weight, etc) on the effect of intervention is not clear, which further increases the difficulty of judging the effect of aerobic exercise intervention.

Although aerobic exercise intervention has shown significant potential in the management of KOA, there are large differences in the form and intensity of aerobic exercise used in different studies, making direct comparison of intervention effects difficult. In addition, the tolerance to exercise varies greatly among different populations, and the applicable intervention program and exercise intensity are difficult to standardize. At the same time, most studies focus on short-term efficacy, and lack of continuous follow-up of long-term intervention effect. The studies on aerobic exercise intervention in KOA are distributed in multiple disciplines and fields, and the data fragmentation is serious, which makes it difficult to fully grasp the whole picture of the research.

Bibliometrics is a method to analyze scientific research trends by quantifying research indicators (such as the number of publications, citation frequency, and core authors), which helps to identify high-impact research, core topics, and frontier trends in the field. Knowledge graph analysis can reveal the research topic, co-occurrence relationship between authors and institutions, cooperation network and topic evolution path by constructing network graph, so as to intuitively show the knowledge structure of this field and explore potential research hotspots and development directions. This study aims to systematically review the current status and hot trends of research on aerobic exercise intervention for KOA in the past decade through bibliometrics and knowledge mapping methods.

The spatial-temporal distribution characteristics of research topics were clarified, the annual publication volume, core authors and research institutions in the field of KOA aerobic exercise intervention in recent years were analyzed, and the global scientific research cooperation network was revealed. Key research hotspots and development directions were identified, and the evolution trend of research focus in this field from traditional exercise intervention to personalized prescription and long-term effect evaluation was analyzed through keyword co-occurrence and cluster analysis. To explore the differences and commonalities of different intervention forms, compare the effects of traditional and modern intervention methods such as water exercise and Tai chi, and find potential optimization directions.^[17,18]

Through the results of this study, we hope to provide a scientific basis and reference for clinicians and researchers in order to provide scientific basis for optimizing exercise intervention strategies for KOA and provide a more comprehensive perspective for future research.

2. Materials and methods

2.1. Data sources and literature search strategies

This study used bibliometric analysis method, based on the literature data of Web of Science (WoS) database, to systematically analyze the related studies of aerobic exercise intervention in patients with KOA. This study did not require approval from an institutional review board because it involved secondary analysis of publicly available data without any human participant involvement. WoS was chosen as the only data source due to its high authority and universality in academia.^[19] As one of the world's leading academic literature databases, WoS covers

high-quality journal articles in multiple disciplines such as natural science, engineering technology, medical health, and social science.^[20] The strict peer review process of the database ensures the academic value of the included literature, and its long-term documentation helps to track the early achievements and the latest progress in the field of research. In addition, the unique citation index function of WoS facilitates the tracking of citation relationships between documents, facilitating co-citation analysis and knowledge map construction.

The search time range of this study was from January 1, 2014 to April 25, 2024. Although the initial time frame covered 2014 to 2015, no eligible articles were retrieved during these 2 years, and the final included publications ranged from 2016 to 2024. Literature retrieval was conducted from the Web of Science Core Collection. To enhance search precision and comprehensiveness, 2 search strategies were designed. The first search formula focused on specific aerobic exercise forms and was constructed as follows: (((((((((TI=(Aerobic exercise)) OR TI=(Tai Chi)) OR TI=(Baduanjin)) OR TI=(Yijinjing)) OR TI=(Wuqinxi)) OR TI=(Bicycle)) OR TI=(Yoga)) OR TI=(Swimming)) OR TI=(Dance)) OR TI=(Pilates)) OR TI=(Take a brisk walk)) OR TI=(Skiing)) OR TI=(Golf)) OR TI=(On foot)) AND TI=(Knee Osteoarthritis)) NOT TI=(Mice)) NOT TI=(Rats). This formula aimed to retrieve detailed studies on various aerobic interventions and excluded animal experimental studies involving mice and rats. The second search formula employed broader subject terms: (TS=(Aerobic exercise)) AND (TS=(Knee Osteoarthritis)), allowing for the inclusion of studies that used different terminologies to describe aerobic exercise interventions. The 2 search formulas were combined using the logic “#1 OR #2,” ensuring both the accuracy and completeness of the retrieved results.

Inclusion criteria were defined as follows: studies involving patients diagnosed with KOA, focusing on aerobic exercise interventions, classified as original research articles or review articles, and published in either English or Chinese. Exclusion criteria included nonhuman studies such as those involving animal experiments (e.g., mice or rats), conference abstracts, editorials, letters, and studies unrelated to KOA or interventions not classified as aerobic exercise. After applying the inclusion and exclusion criteria, a total of 273 eligible studies were included for quantitative analysis. The retrieval process is illustrated in Figure 1.

2.2. Statistical methods

In the process of data analysis, 4 knowledge graph analysis softwares, including CiteSpace, VOSviewer, COOC14.9, and SCImago Graphica, were used in this study to achieve multidimensional and refined analysis. COOC14.9 was used to calculate the annual number of publications, the cumulative number of publications, authors, institutions, keywords, national publication frequency, annual keyword attention change, annual keyword growth rate change, and time-weighted keyword hot spot change, and draw rose chart, radar chart, pie chart, and cumulative line chart. COOC14.9 has diversified co-occurrence analysis functions and can identify the links between keywords, authors and institutions through co-occurrence networks, which is a suitable analysis tool to show annual changes and topic distribution.^[21,22] SCImago Graphica was used for national cooperation analysis to generate national Geographic cooperation map, cooperation curve and chord chart, and reveal the global cooperation mode by analyzing the scientific research cooperation of different countries in a certain field. As an international cooperation analysis platform for academic publications, SCImago Graphica is good at geographical distribution visualization, which is suitable for presenting the cooperation relationship between countries and its evolution.^[23] VOSviewer is mainly used to analyze keyword co-occurrence, heat map of national papers, publication time and cooperation time between countries. It has advantages in the analysis of keyword

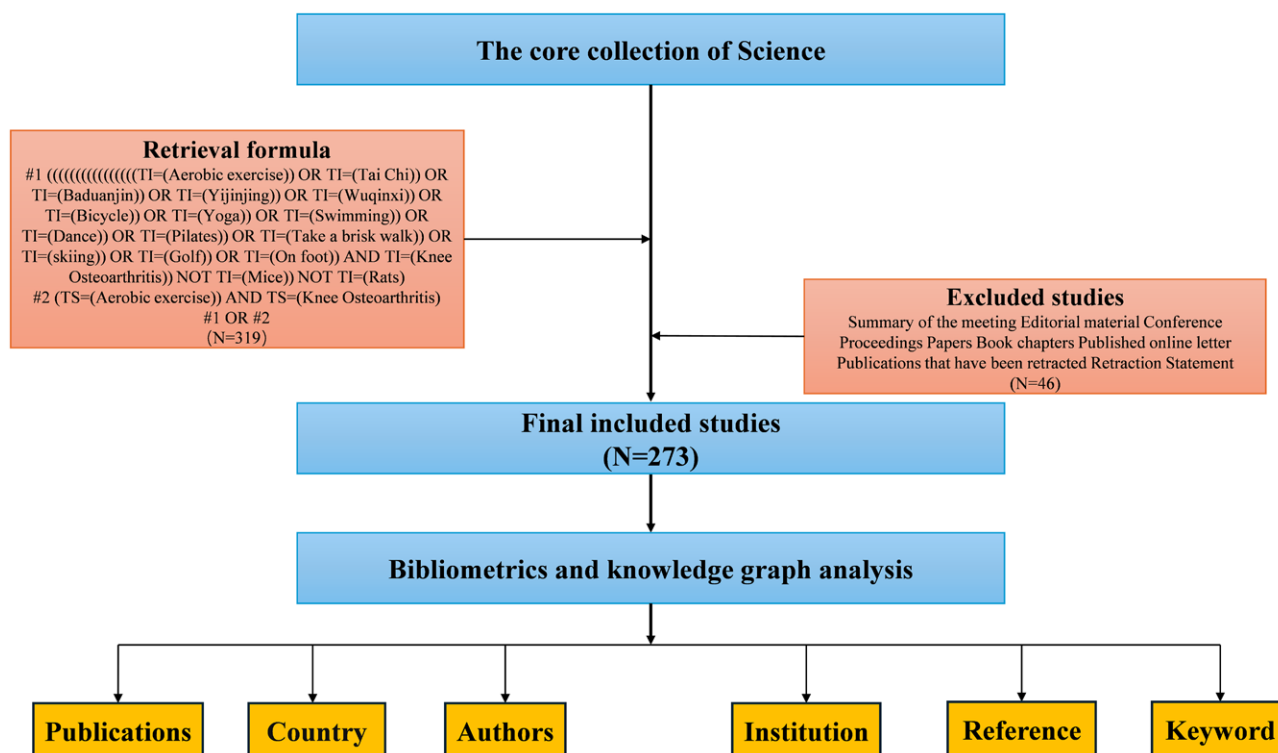


Figure 1. Flow chart for screening publications.

co-occurrence and cooperation time trend, which can intuitively show the evolution of research topics and the distribution of academic cooperation between countries.^[24] CiteSpace was used to analyze author co-occurrence, institution co-occurrence, co-citation, keyword clustering, journal double graph overlay, and keyword timeline analysis. CiteSpace is designed for scientific literature analysis. It is good at timeline and co-citation analysis, which can reveal the development context and evolution trend of research topics.^[25]

The time slice of all software was set to 1 year to ensure the fineness of the analysis and to capture the annual research hotspots and trend evolution. The selection method of nodes is *g*-index. *g*-Index has strong flexibility in evaluating academic influence, and can identify high frequency nodes and low frequency nodes at the same time, so as to retain the potential hotspots in research. The network pruning algorithm is selected according to the different analysis content. Pathfinder algorithm is used for co-occurrence network, which can simplify the network structure and retain the core nodes and main relationships. The minimum spanning tree algorithm is used to remove redundant connections through the minimum spanning tree, making the graph more concise and easy to read.^[26] Cluster analysis uses *k*-means clustering method,^[27] which is suitable for data grouping based on similarity and can help to identify the main research themes. The clustering results are evaluated by a combination of *Q*-values, which measure modularity, that is, the independence of clusters, and *S*-values, which measure the closeness within clusters. When both the *q* and *s* values are high, the clustering effect is good, indicating that the topic clustering is both clear and internally consistent in the network structure.

3. Results

3.1. Analysis of the annual number of publications and the cumulative number of publications

From 2016 to 2024, the annual number of publications on aerobic exercise intervention in the field of KOA showed a certain

fluctuation trend, but maintained a stable growth as a whole. The left of Figure 2 shows the annual change in the number of publications, and the right shows the continuous growth of the cumulative number of publications.

3.1.1. Analysis of annual published papers. From 2016 to 2018, the annual number of published articles increased gradually, from 25 articles in 2016 to 31 articles in 2018. There was a slight drop to 28 papers in 2019, but then the growth resumed in 2020 and 2021, reaching 32 and 34 papers, respectively, indicating that the research popularity has increased. In 2022, the annual number of articles published reached 39, which was the highest value in this period of time, reflecting that the academic research attention to aerobic exercise intervention for KOA has reached the peak. Although the number of papers published in 2023 and 2024 will decline, with 30 and 28 papers respectively, the overall level is still relatively stable, showing that the research heat in this field is still high.

3.1.2. Cumulative number of publications analysis. The cumulative number of publications in the right graph shows a clear linear growth trend. Starting from 25 articles in 2016, the cumulative number of articles published increased year by year, reaching 82 articles in 2018, 110 articles in 2020, 176 articles in 2022, and 273 articles in 2024. This steady upward trend shows that in the past 8 years, the academic community has gradually paid more attention to aerobic exercise intervention for KOA, and the cumulative research results have also been increasing, which has laid a solid theoretical and practical foundation for this field.

3.2. Country or region analysis

3.2.1. Analysis of the number of publications. From the perspective of the number of publications in each country or region, the United States, China, and Australia dominate the research in the field of aerobic exercise intervention for KOA. The United States topped the list with 69 articles, reflecting its highest

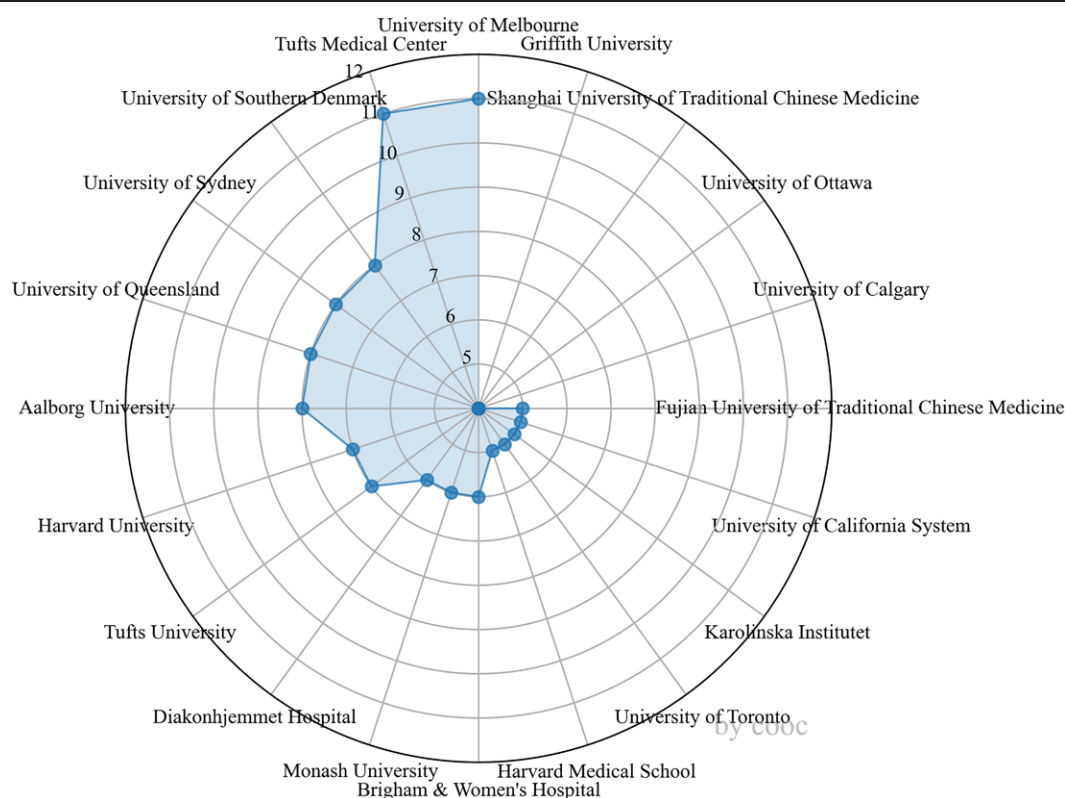


Figure 2. Line chart of annual publications and cumulative publications.

Table 1

Top 10 countries by number of publications.

Country	Number of publications
United States	69
China	61
Australia	39
Canada	34
Brazil	20
Denmark	18
England	18
Netherlands	15
Belgium	13
Japan	13

research resources and academic investment in this field. China followed closely, with a total of 61 papers, showing a gradual increase in scientific research output in recent years. Australia ranked third with 39 articles, reflecting the country's strength in health and exercise science. Canada (34 articles), the United Kingdom (18 articles), Denmark (18 articles), and the Netherlands (15 articles) also had high research activity in this field.

From the perspective of the overall distribution of publications, the studies were mainly concentrated in developed countries or regions in Europe, America, and the Asia-Pacific region. This regional difference may be related to the investment in medical and sports science research in these countries and the adequacy of the related infrastructure. In general, international attention to aerobic exercise intervention for KOA has increased year by year, and this trend has also promoted the continuous development of global research to a certain extent. As shown in Table 1 and Figure 3C.

3.2.2. Analysis of cooperation. In the cooperation network between countries or regions, the United States, China, and

Australia not only rank at the forefront in terms of the number of publications, but also show high connection strength in international cooperation. The results of SCImago Graphica analysis (Fig. 3A) show that the scientific research cooperation between developed countries is closer, such as the United States and Canada, Australia, the United Kingdom, and China have frequent academic exchanges. The total link strength of the United States peaked, demonstrating its position as a center of international cooperation in this field. China is also active in international cooperation and has established a solid cooperation network with Australia, the UK, and Brazil.

The breadth of cooperation among different countries can be observed through the geographic cooperation and chord maps drawn. As shown in the cooperation graph (Figs. 3B and 4), the United States, Australia, and China form the core cluster of cooperation and have a strong influence on the global scale. European countries such as Denmark, the Netherlands, and the United Kingdom form another sub-core region with closer research cooperation. This cooperative relationship reflects the globalization trend in this field, especially in terms of technology sharing, resource sharing and academic exchange. International cooperation provides important support for promoting the scientific research progress of KOA aerobic exercise intervention.

3.2.3. Cooperation time analysis. Figure 5 shows the cooperation network and popularity of various countries in KOA research from 2019 to 2021. The data across the 3 years show the trends and intensity of the collaboration among the major countries, indicated by the time points of the major collaboration in each country by different colors.

The color of the nodes in the figure gradually changes from blue to yellow and red, indicating the main time points in which countries cooperate in different years. Blue represents the early collaboration (2019), green to yellow represents the gradually deeper stage of collaboration (2020), and red represents the

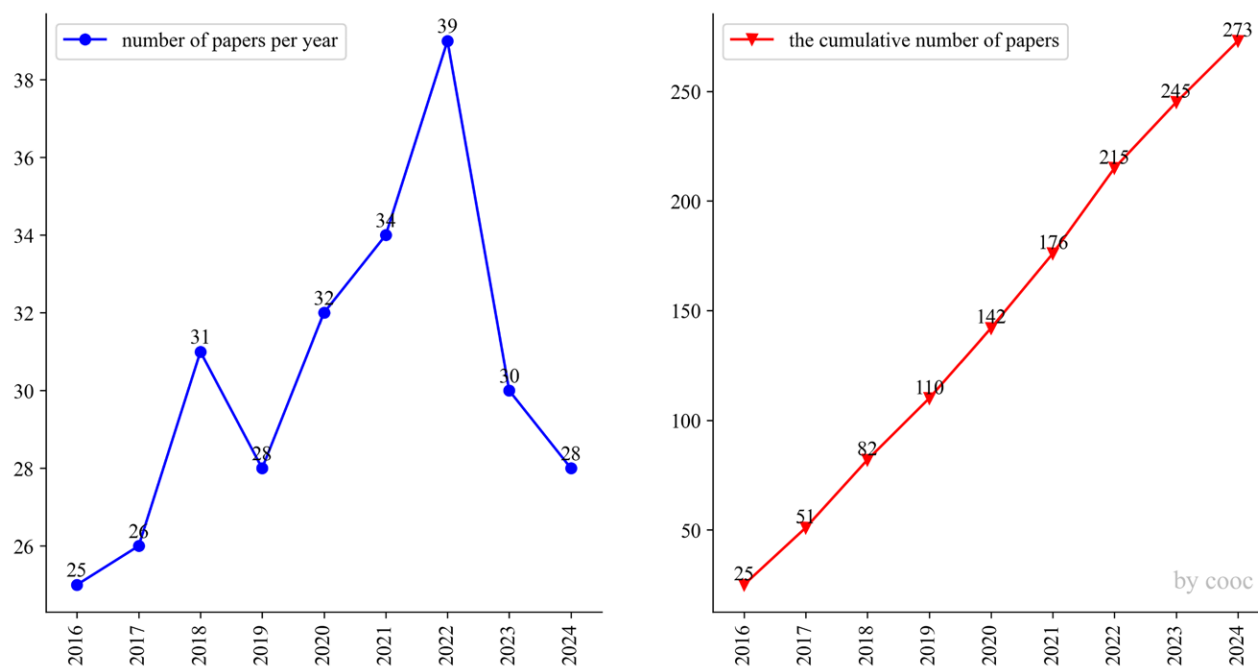


Figure 3. Country synthesis analysis. (A) A scatter plot displays country clustering by cooperation patterns. Node colors denote clusters (green for cluster 1, light blue for cluster 2, orange for cluster 3, purple for cluster 4). Lines between nodes indicate inter-country cooperative relationships. (B) A network diagram illustrates international country cooperation. Node sizes reflect publication counts (larger nodes mean more publications, ranging from 30–90). Node and line colors, gradient from cyan to dark blue, represent cooperation strength (ranging from 1–49). (C) A chord diagram shows cooperative relationships among different countries. Each sector corresponds to a country.

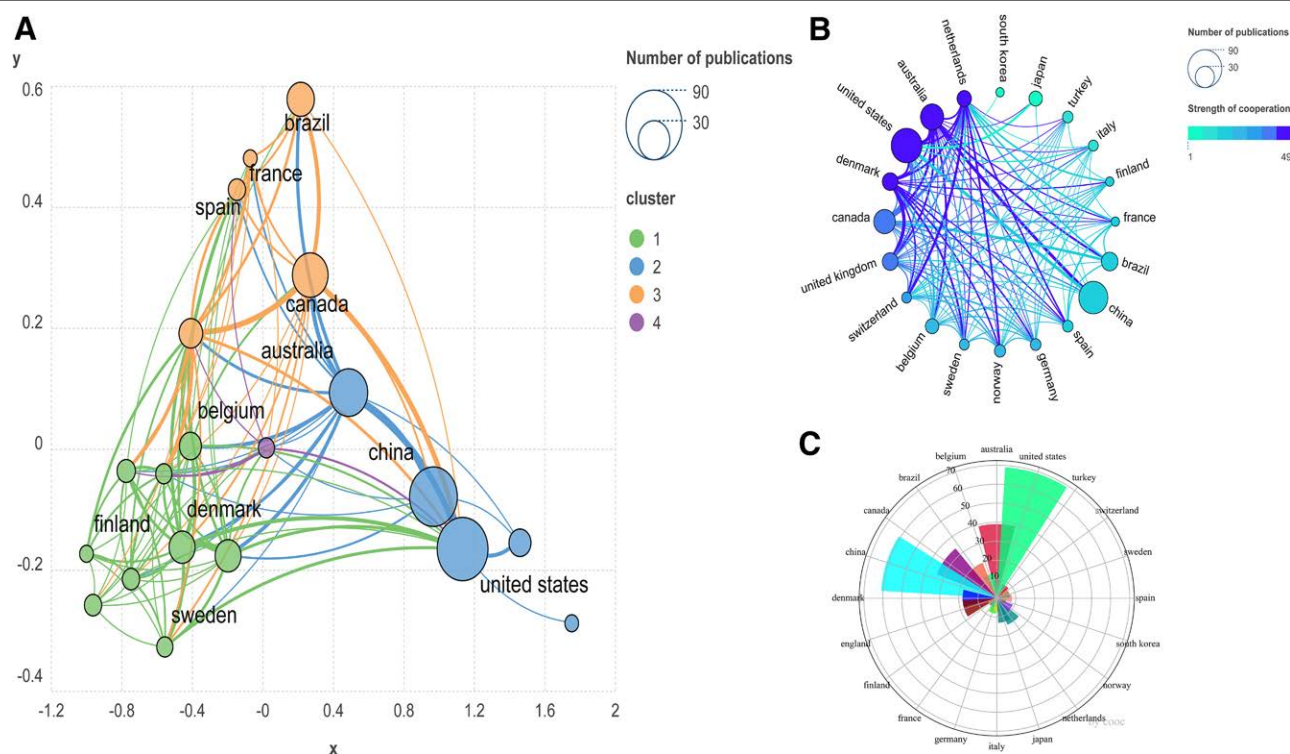


Figure 4. Publication and cooperation in high-yield countries. Top 20 countries in terms of publication volume.

more recent collaboration (2021). This reflects the variation in the distribution of activity and time spent on research in different countries.

The United States is the most important hub for collaboration in KOA research, with the largest number of nodes,

indicating its dominance in international collaboration. Its cooperation time is mainly concentrated in 2020 to 2021, which also corresponds to the color of green and yellow. China also has larger nodes in the figure, showing its important position in the field of KOA. The color gradually changes to red,

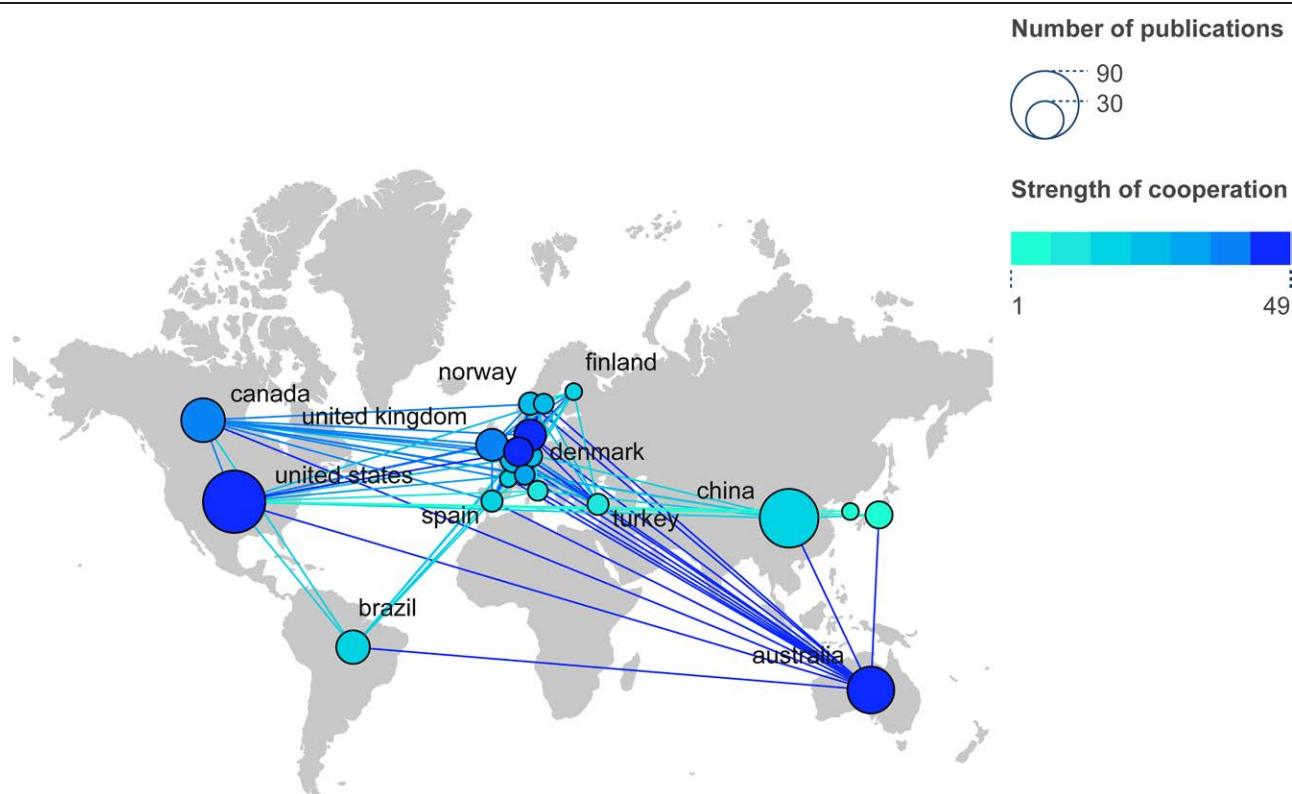


Figure 5. National publication time and heat map. (A) A scatter plot presents cooperative relationships among countries. Node sizes reflect the number of publications of respective countries. Lines connecting nodes signify inter-country cooperative ties, with line thickness denoting cooperation closeness. Node colors, gradient from blue to red, indicate publication time from 2019 to 2021. (B) A heat map depicts publication heat across different countries. Color intensity, ranging from blue (low heat) to red (high heat), shows publication heat levels. Distinct regions in the heat map correspond to different countries, and color distribution within regions illustrates each country's publication heat spread.

indicating that the collaboration is concentrated in 2021. This indicates that China has invested more scientific research resources in this field in recent years, and its cooperation with the United States, Australia, and other countries has been particularly active in recent years. The color of the Australian node is between green and yellow, indicating its more frequent cooperation around 2020. Australia maintains close cooperation with the United States and China, and has relatively solid cooperation with several European countries (such as the United Kingdom, the Netherlands, Denmark, etc). This shows that Australia is a bridge connecting major continents in scientific research cooperation in these 3 years. European countries such as the United Kingdom, the Netherlands, and Denmark maintain active cooperation among themselves, and there are many connections between nodes, indicating close connections within the region. In addition, these countries have strong cooperation ties with the United States and Australia. A large number of collaborations occurred between 2019 and 2021, and the international cooperation network showed obvious dynamic changes. The pandemic has led to an increase in remote collaboration and increased research intensity in some countries. Analysis of this time period provides a valuable perspective to explore these changes. From the perspective of time, the cooperation of European countries gradually deepened from 2019 to 2021, and the color change shows their gradually increasing participation in scientific research.

In 2019, the cooperation was mainly led by the United States and several European countries. The cooperation scale was small and the color was mainly blue, indicating that the cooperation was just starting to take off. At this stage, researchers focused more on intra-regional cooperation, and international cooperation had not been carried out on a large-scale. In 2020,

the cooperation began to deepen and expand, with a significant increase in the frequency and breadth of cooperation. The nodes in the United States and Australia are in green, indicating that they have increased their efforts to cooperate in 2020, especially in research links with China and Europe, where research investment and data sharing have been further deepened. After entering 2021, the cooperation node gradually turns red, which represents the increase of cooperation intensity and activity. The Chinese node is particularly evident in this year, showing that China has become a new and important force in KOA research and has begun to participate more in international cooperation. The United States and Australia have also reached a new high-level of cooperation, especially with European countries.

3.3. Analysis of the number of publications and co-occurrence of authors

As shown in Table 2, in the study of aerobic exercise intervention in the field of KOA, Bennell, Kim L and Brosseau, Lucie are the core authors in this field, with high academic influence and a large number of publications. According to the author co-occurrence diagram (Fig. 6) and Rose diagram (Fig. 7) generated by CiteSpace, Bennell, Kim L, Brosseau and Lucie are 2 prominent research centers, leading different research networks and forming obvious cooperation clusters respectively. Diverse research directions and academic achievements in this field are presented.

Bennell, Kim L is one of the major contributors to this field, with research focused on the functional improvement and pain relief of aerobic exercise in patients with KOA. Bennell's research group was relatively diverse, with extensive and

dispersed collaborations, exhibiting an open academic network structure. Its close collaboration with Collins, Natalie J and Button, Kate and others has facilitated the multi-angle evaluation of the effects of aerobic exercise interventions. According to the co-occurrence map of the authors, the research network of Bennell and Kim L includes multiple subclusters, and these subclusters are closely related to each other, showing high intensity of collaborative research. This network structure suggests that Bennell has advanced the scientific development of aerobic exercise interventions in the management of KOA through collaboration across teams. His team's research topics include the physiological mechanisms of aerobic exercise interventions, comparative studies of different forms of exercise, and long-term patient management options, covering the key topics in this field.

Another important researcher, Brosseau and Lucie, focused on systematic evaluation and evidence-based medicine, and most of his studies were systematic reviews and large-scale clinical trials, aiming to summarize and verify the clinical application effect

of aerobic exercise in KOA. The network of Brosseau and his team was relatively concentrated, and the key members included Desjardins, Brigit, Mclean, Linda and Van der Esch, Martin, etc, who played an important role in the summary of clinical evidence and the construction of evidence-based framework for aerobic exercise intervention. Brosseau's research team closely integrated clinical trial data and systematic literature review to provide effective evidence-based support for KOA patients by comparing different types of aerobic exercise and tracking the long-term efficacy. In addition, the group's research is broad, covering the applicability of multiple forms of exercise (e.g., walking, tai chi, swimming, etc), and it focuses on interdisciplinary collaboration, bringing together biomechanics, exercise rehabilitation, and clinical epidemiology.

In addition to Bennell and Brosseau, Herzog, Walter, Wang, Chenchen, Simic, Milena and Kenny, Glen P, and other scholars are also active researchers in this field. The rose chart data showed that these authors had a high number of publications and a strong academic influence in their respective research directions. For example, Herzog and Walter had in-depth research on sports biomechanics, which provided important scientific basis for understanding the physiological effects of different forms of aerobic exercise on knee joints. Wang and Chenchen focused on patient self-management and formulation of exercise prescription, and the research content had strong practical guidance. The study of Simic, Milena covered the individualized adjustment of exercise intensity and frequency, providing the basis for personalized exercise intervention programs.

From the co-occurrence map cooperation network, it can be seen that the aerobic exercise intervention research in the field of KOA shows a relatively concentrated cooperative mode. Bennell and Brosseau lead 2 large clusters, respectively, and integrate academic resources in the field through multilevel collaboration. The Bennell cluster has the characteristics of cross-team and interdisciplinary research, covering many fields

Table 2
Top 10 authors with the largest number of publications.

Author	Number of publications
Bennell, Kim L.	6
Bennell, Kim	4
Herzog, Walter	4
Brosseau, Lucie	3
Wang, Chenchen	3
Price, Lori Lyn	3
Niedermann, Karin	2
Aburub, Ala'	2
Letafatkar, Amir	2
Van der esch, Martin	2

CiteSpace, v. 5.4.R1 (64-bit) Advanced
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WOS: D:\WOSData
Timespan: 2016-2024 (Slice Length=1)
Selection Criteria: g-index (k=25), LRF=2.5, L/N=10, LBY=5, q=1.0
Network: N=265, E=373 (Density=0.0107)
Largest 1 CCs: 41 (15%)
Nodes Labeled: 1.0%
Pruning: Pathfinder
Excluded:

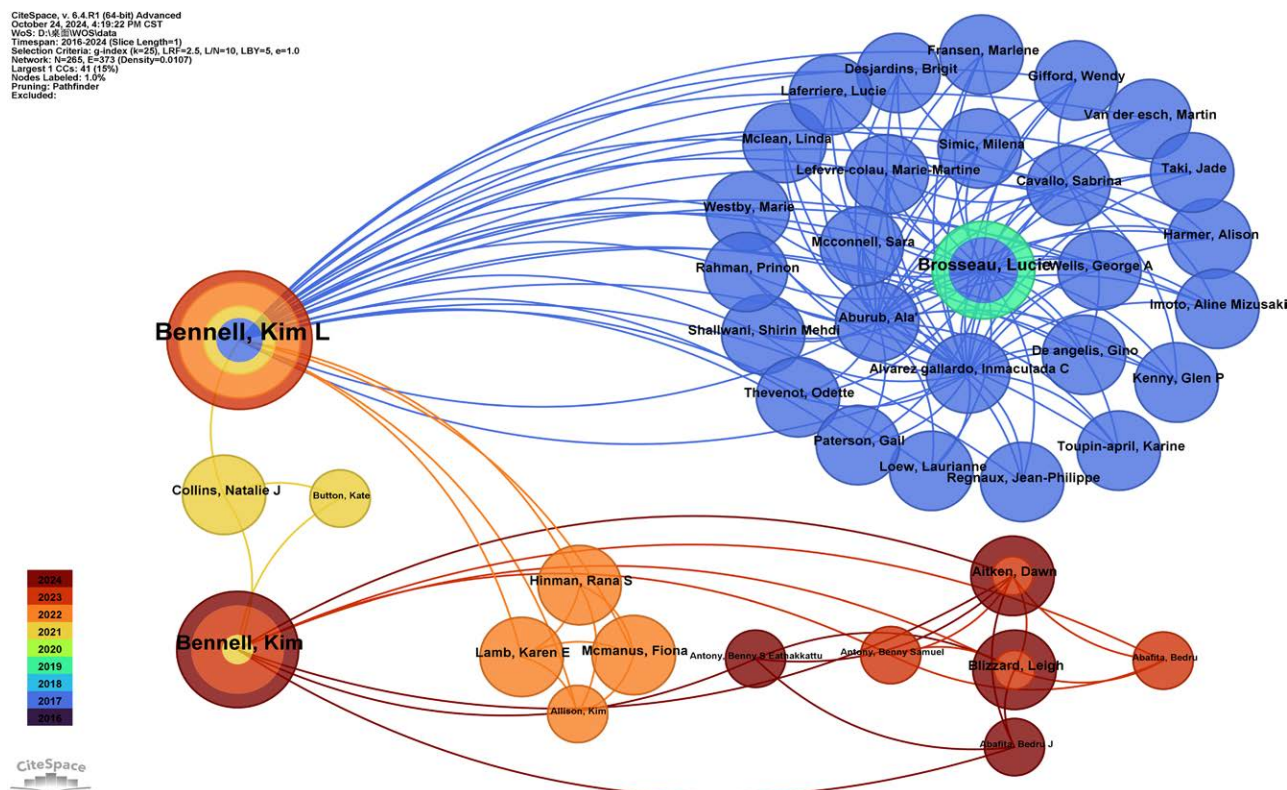


Figure 6. Author co-occurrence diagram.

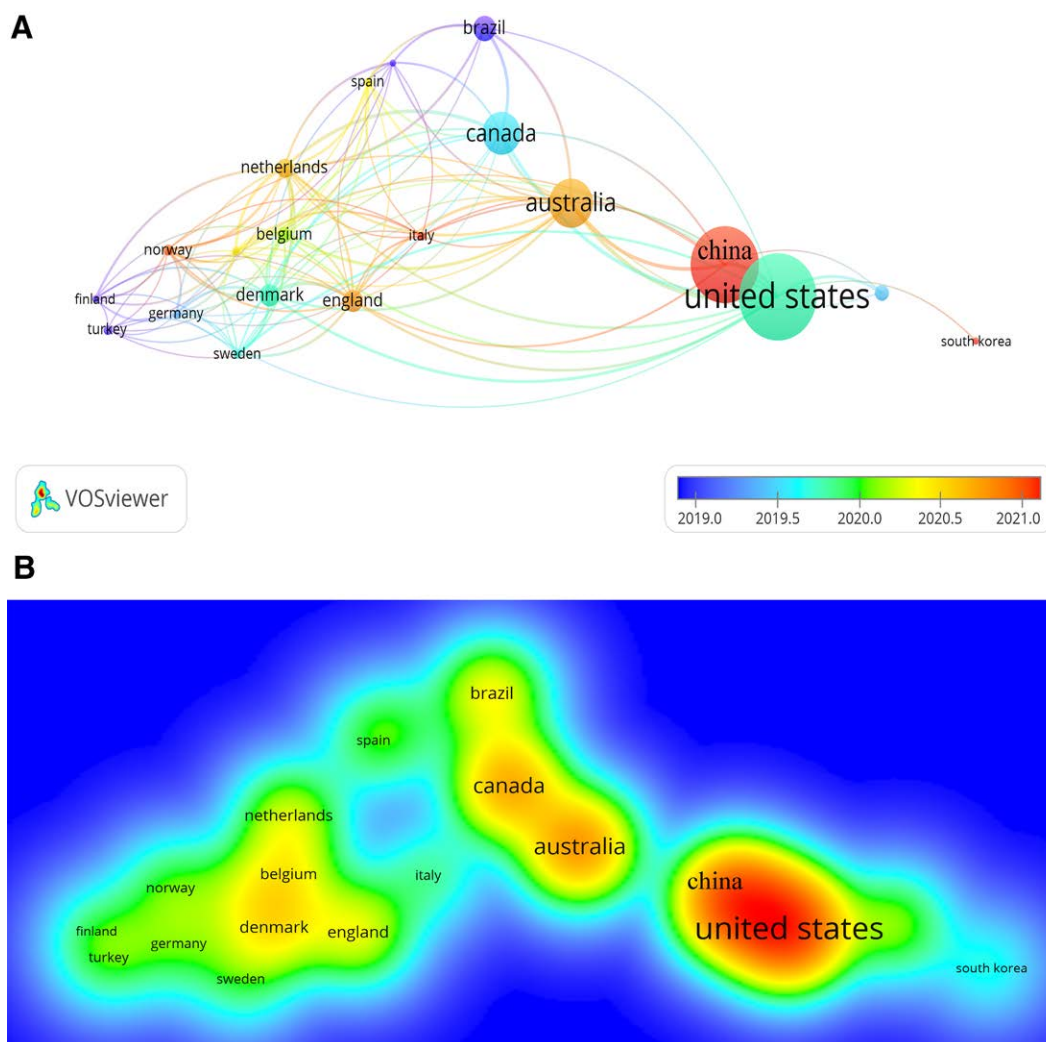


Figure 7. Rose chart of the author's number of publications.

Table 3

Top 10 institutions by number of publications.

Institution	Number of publications
University of Melbourne	11
Tufts Medical Center	11
University of Southern Denmark	8
University of Sydney	8
University of Queensland	8
Aalborg University	8
Harvard University	7
Tufts University	7
Diakonhjemmet Hospital	6
Monash University	6

such as physiology, rehabilitation and public health. However, Brosseau cluster pays more attention to systematic evaluation and summary of clinical evidence, showing its advantages in evidence-based medicine and clinical practice.

This cluster distribution and cooperation pattern shows the trend of diversification and internationalization of research within the field. The nodes in the cooperative network represent different academic teams, and each team jointly promotes the academic development of aerobic exercise intervention in the treatment of KOA through different research topics and

methods. In addition, the cooperative relationship between the authors shows a certain academic clustering effect, for example, the research in the Bennell cluster focuses on empirical research, while the Brosseau cluster is more inclined to review and systematic analysis. Through this diversified research approach, different teams play complementary roles in the body of knowledge on KOA aerobic exercise intervention research, driving development and innovation in the field.

3.4. Analysis of the number of publications and co-occurrence of institutions

In the study of aerobic exercise intervention in KOA, the number of publications and cooperative relationships of different academic institutions show the trend of global and academic cooperation in this field. Table 3 and radar chart (Fig. 8) show the publication frequency of different institutions, among which the University of Southern Denmark and the University of Melbourne rank in the top, and the number of publications is significantly higher than that of other institutions. The number of papers published by Shanghai University of Traditional Chinese Medicine was also prominent, indicating that the status of traditional Chinese medicine in international rehabilitation medicine research was gradually improved. According to the co-occurrence map, radar map and the publication data of the institution, You can see the University of Melbourne and University of

Queensland in Australia, the University of Southern Denmark in Denmark, and the Shanghai University in China. Institutions such as Traditional Chinese Medicine and Harvard University in the United States are at the core of this field, reflecting the important contributions of these institutions in the research of KOA aerobic exercise intervention. The University of Melbourne and the University of Queensland are 2 important research institutions in Australia, especially in the empirical research and effect evaluation of aerobic exercise intervention. Researchers from both universities work closely together on basic and applied research in the fields of sports medicine and rehabilitation.

The University of Southern Denmark has a prominent academic influence in Europe. The research team of the University focuses on the effects of different forms of aerobic exercise on joint function and quality of life of KOA patients, and the

research content covers sports biomechanics and exercise prescription optimization.

Shanghai University of Traditional Chinese Medicine represents China's innovative research in the integration of traditional exercise and modern rehabilitation medicine. The research of this institution mainly focuses on the effects of traditional aerobic exercises such as Tai Chi and Baduanjin on KOA intervention, and is dedicated to combining traditional medicine with modern rehabilitation methods. Harvard University, as a well-known research institution in the United States, has a leading advantage in clinical trials and evidence-based medicine of KOA aerobic exercise intervention. The Harvard team focused specifically on the evaluation of long-term intervention effects and worked to develop personalized exercise intervention programs.

According to the co-occurrence map of institutional cooperation and the cooperative relationship between institutions, the major research institutions in the world have formed several cooperative clusters (Fig. 9). University of Melbourne has close cooperation with institutions such as University of Southern Denmark, Aalborg University and University of Sydney. Formed a network of cooperation with Australia and Europe as the core. Together, these institutions have promoted research on aerobic exercise interventions based on biomechanics and rehabilitation medicine. The University of Queensland has international cooperation with Deakin University, University of Bristol, University of Toronto and other institutions. In particular, it has played an important role in the study of intervention effects and functional recovery of patients. As Sweden's representative institution, Karolinska Institutet has joined forces with other research centers in Europe, such as Zurich University of Applied Sciences and Ruhr University Bochum. The common focus is on the physiological mechanisms and mental health effects of exercise interventions.

3.5. Literature and journal citation analysis

3.5.1. Citation analysis of high-impact literature. The analysis of high-impact literature is to evaluate the literature with high citation frequency and intensive citation in a specific time period, which has high recognition and academic value in

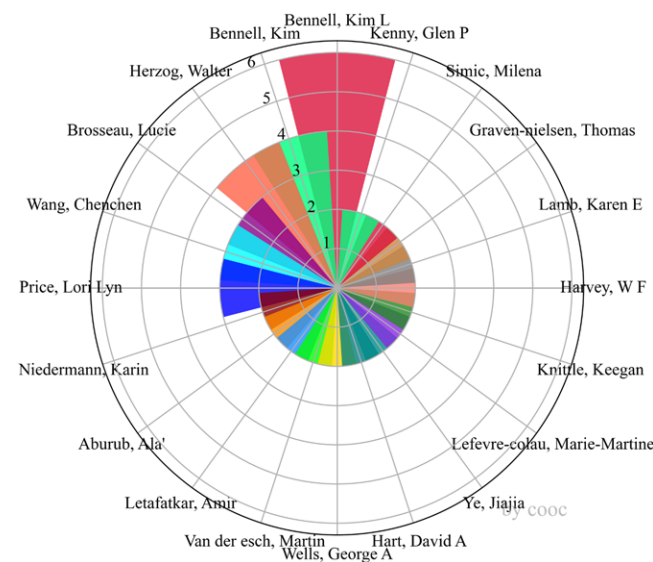


Figure 8. Radar chart of the institution.

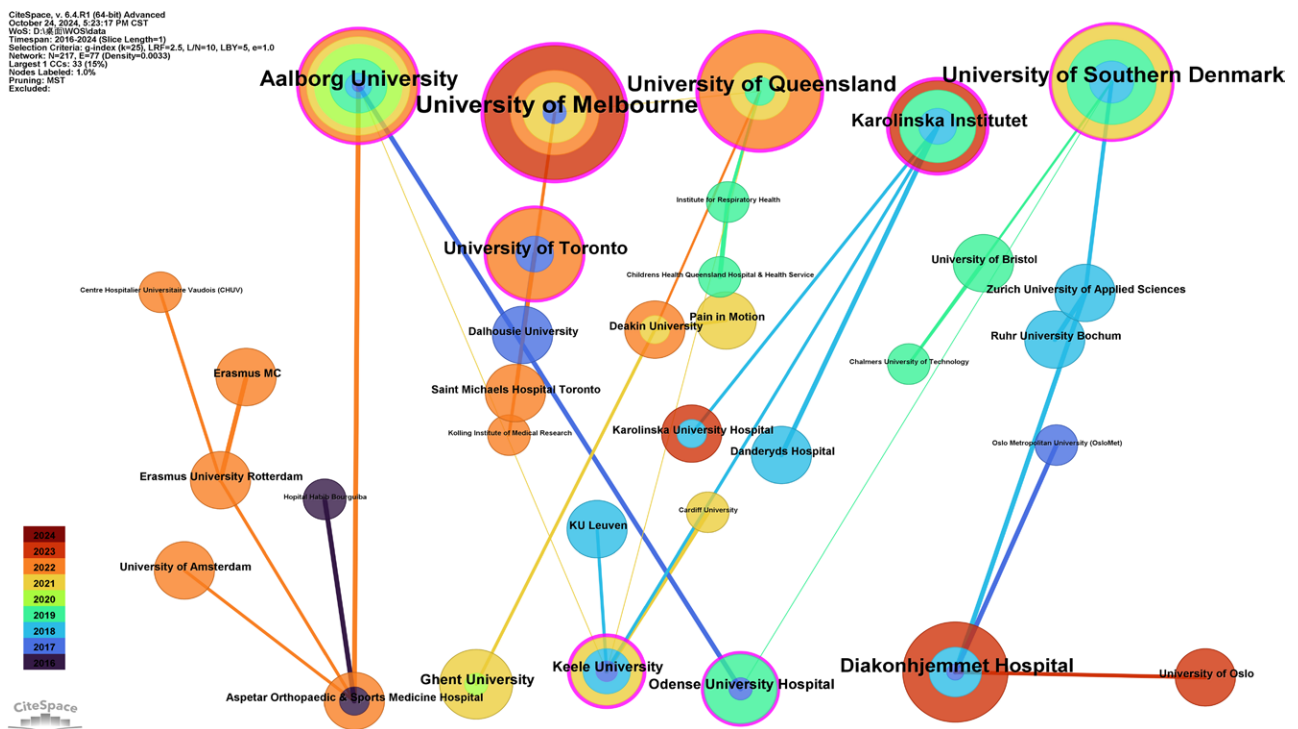


Figure 9. Institutional co-occurrence map.

Top 20 References with the Strongest Citation Bursts

References		Year Strength Begin End 2016 - 2024			
McAlindon TE, 2014, OSTEOARTHR CARTILAGE, V22, P363, DOI 10.1016/j.joca.2014.01.003, DOI	2014	5.53	2016	2019	
Fernandes L, 2013, ANN RHEUM DIS, V72, P1125, DOI 10.1136/annrheumdis-2012-202745, DOI	2013	4.43	2016	2018	
Uthman OA, 2013, BMJ-BRIT MED J, V347, P0, DOI 10.1136/bmj.f5555, DOI	2013	3.62	2016	2018	
Hochberg MC, 2012, ARTHRIT CARE RES, V64, P465, DOI 10.1002/acr.21596, DOI	2012	3.18	2016	2017	
Tanaka R, 2013, CLIN REHABIL, V27, P1059, DOI 10.1177/0269215513488898, DOI	2013	2.65	2016	2017	
Fransen M, 2015, COCHRANE DB SYST REV, V0, P0, DOI 10.1002/14651858.CD004376.pub3, DOI	2015	2.63	2016	2020	
Juhl C, 2014, ARTHRITIS RHEUMATOL, V66, P622, DOI 10.1002/art.38290, DOI	2014	5.1	2017	2019	
Hoffmann TC, 2014, BMJ-BRIT MED J, V348, P0, DOI 10.1055/s-0041-111066, 10.1136/bmj.g1687, DOI	2014	2.53	2017	2019	
Bartholdy C, 2017, SEMIN ARTHRITIS RHEU, V47, P9, DOI 10.1016/j.semarthrit.2017.03.007, DOI	2017	2.24	2018	2019	
Samut G, 2015, MOD RHEUMATOL, V25, P919, DOI 10.3109/14397595.2015.1038425, DOI	2015	2.3	2019	2020	
Ghandali E, 2017, J BODYW MOV THER, V21, P594, DOI 10.1016/j.jbmt.2016.09.001, DOI	2017	3.89	2020	2021	
Zhu QG, 2016, J SPORT HEALTH SCI, V5, P297, DOI 10.1016/j.jsbs.2016.02.003, DOI	2016	3.02	2020	2021	
Bartels EM, 2016, COCHRANE DB SYST REV, V0, P0, DOI 10.1002/14651858.CD005523.pub3, DOI	2016	2.58	2020	2021	
Brosseau L, 2017, CLIN REHABIL, V31, P612, DOI 10.1177/0269215517691085, DOI	2017	2.37	2020	2022	
Bannuru RR, 2019, OSTEOARTHR CARTILAGE, V27, P1578, DOI 10.1016/j.joca.2019.06.011, DOI	2019	6.41	2021	2024	
Hunter DJ, 2019, LANCET, V393, P1745, DOI 10.1016/S0140-6736(19)30417-9, DOI	2019	3.03	2021	2024	
Lü JJ, 2017, SLEEP MED, V33, P70, DOI 10.1016/j.sleep.2016.12.024, DOI	2017	2.3	2021	2022	
Kolasinski SL, 2020, ARTHRIT CARE RES, V72, P149, DOI 10.1002/acr.24131, 10.1002/art.41142, DOI	2020	7.8	2022	2024	
Sharma L, 2021, NEW ENGL J MED, V384, P51, DOI 10.1056/NEJMcp1903768, DOI	2021	3.75	2022	2024	
Page MJ, 2021, BMJ-BRIT MED J, V372, P0, DOI 10.1016/j.ijisu.2021.105906, 10.1136/bmj.n160, 10.1136/bmj.n71, DOI	2021	3.37	2022	2024	

Figure 10. Citation outburst statistics chart.

the field. Through the presentation of Figures 10 and 11, we can clearly see which literatures are the hot spots and core reference points in the research field from 2014 to 2024.

3.5.1.1. Citation emergent literature detection. “Top 20 References with the Strongest Citation Bursts” in Figure 10 lists the top 20 references with the strongest citation bursts, reflecting that these papers have received particularly high academic attention in some time periods.

For example, the citation strength of McAlindon et al (2014) paper from 2016 to 2019 was 5.53, reflecting that this paper has a significant impact on the field of KOA in these years. Its research may involve innovative therapeutic methods, important clinical findings, or new theoretical frameworks, making scholars frequently cited during this period.

The citation intensity of Bannuru et al (2019) was 6.41 from 2021 to 2024, which represents that this article has been widely cited by scholars in recent years. This may be because its research results have brought new directions or solved important problems in the field of KOA.

Among the top 20 highly cited burst articles, most of them were concentrated between 2016 and 2024, indicating that aerobic exercise intervention research on KOA has received sustained and extensive academic attention in the past 10 years.

3.5.1.2. Co-occurrence analysis of cited literature. Citation co-occurrence analysis is an effective method to identify the core research basis and hotspots in an academic field. By analyzing the co-occurrence relationship between the cited literatures, the important theoretical framework and the evolution process of research topics in the field can be revealed. The structure of the co-cited literature network shown in Figure 11B can intuitively reflect the influence of a particular literature in the research field and its degree of association with other literature.

As can be seen in Figure 11B, the size of nodes is proportional to their citation frequency, and larger nodes represent the high citation frequency of the literature in the selected time range, indicating its importance and influence in the field. For example, the nodes of Bannuru R.R. (2019) and Kolasinski S.L. (2020) were significantly larger, indicating that these 2 papers are the core papers in the field of KOA related research. They may involve the latest reviews or key research advances in

therapeutic interventions for KOA, providing theoretical support and practical guidance for subsequent academic research.

The color of the node indicates the time range in which the literature was cited, with dark colors (e.g., blue) representing earlier high-impact literature and light colors (e.g., red) representing new literature with high citation frequency in recent years. In the figure, the literature nodes such as Uthman O.A. (2013) and Juhl C. (2014) are shown in dark blue, indicating that these are the key literatures in the early period, which laid important theories for the foundation of the field. Nodes such as Sharma L. (2021) and Page M.J. (2021) are in light color, indicating that these literatures have gradually increased their influence on research in recent years.

The lines between the nodes indicate the co-citation relationship between the articles, and the thicker the lines, the more times they have been co-cited in the same reference. For example, the connection between Bannuru R.R. (2019) and Fransen M. (2015) is relatively robust, indicating that these 2 articles have been co-cited in many subsequent studies, which may have important theoretical contributions or experimental basis in similar research directions. The literatures with high citation frequency are usually review articles or guidelines in the field, which have high citation value and become the theoretical basis for other researchers in related research. For example, Bannuru R.R. (2019) and Fransen M. (2015), as literatures with large nodes, indicate that they may play an important role in the research of nondrug treatment or intervention therapy for KOA, and have been widely cited as theoretical basis and reference for subsequent research.

3.5.2. Citation networks among different disciplines and their evolution analysis. Figure 11A illustrates the structure of the citation network across different subject areas in the study of aerobic exercise interventions for KOA. Through the knowledge graph, the cross-disciplinary cooperation of various disciplines can be intuitively seen, reflecting the multidisciplinary characteristics of research in this field.

The left side mainly includes basic science fields such as molecular biology, immunology and clinical medicine, and the right side covers applied disciplines such as health care, psychology, education, and social science. These disciplines have carried out various studies on the effects of aerobic exercise

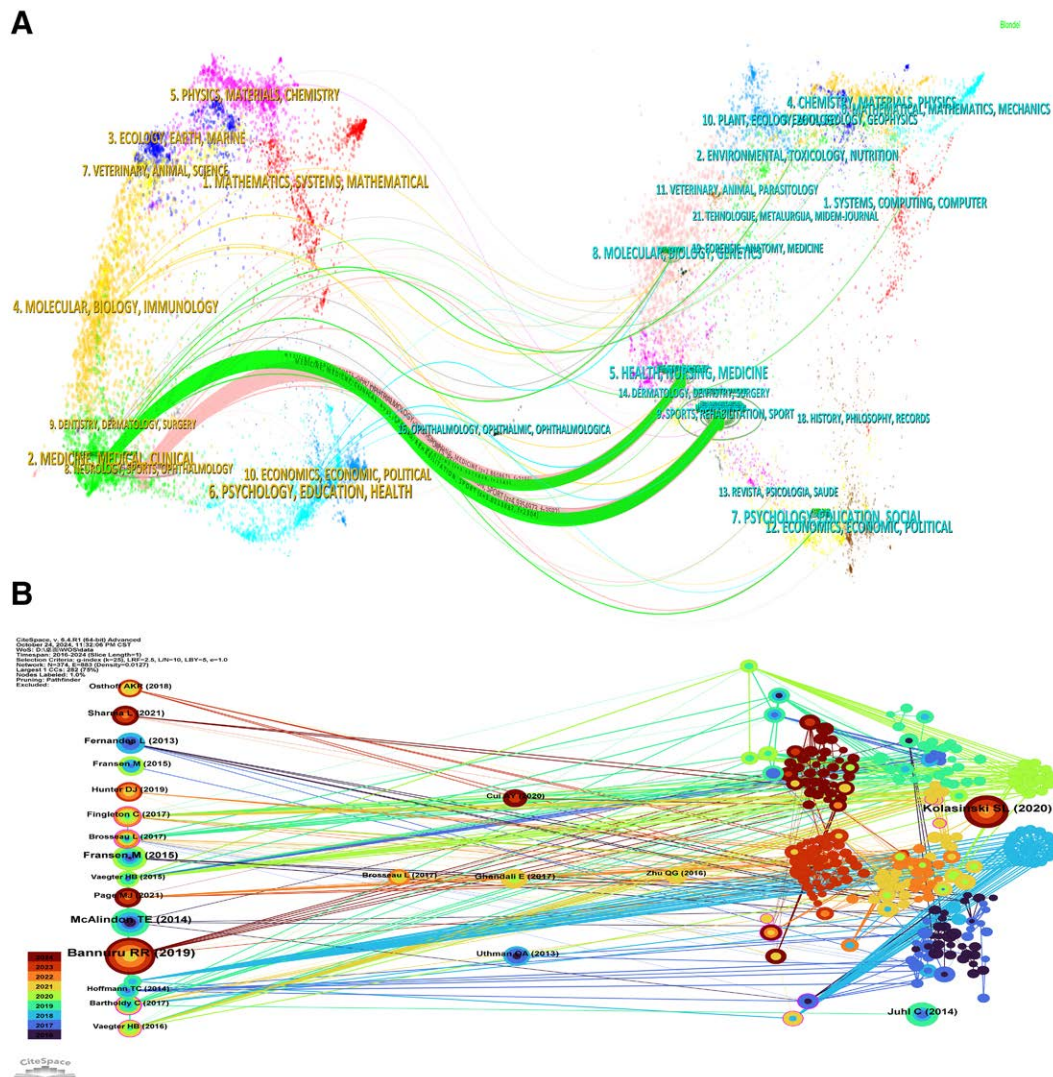


Figure 11. The co-occurrence network and evolution trend graph are cited. (A) A co-occurrence network diagram of different research fields. Distinct colors represent separate research fields. Node sizes correspond to the occurrence frequency of the fields (larger nodes indicate higher frequency). Lines between nodes represent the co-occurrence relationships among different research fields, and line thickness indicates the strength of such relationships (thicker lines mean stronger co-occurrence). (B) An evolution trend graph of research in different fields. Diverse colors represent specific research fields. Node sizes are determined by the number of citations and research influence (larger nodes indicate higher citations and greater influence). Lines between nodes represent the evolution or inheritance relationships among different research fields, with line thickness reflecting the strength of these relationships (thicker lines indicate more significant evolutionary connections). Nodes are marked with research topics and publication years, clearly demonstrating the temporal development trend of the research.

intervention, including biological mechanisms, clinical effects, and psychological support.

It can be seen that different disciplines are related to each other through different colored paths such as the green line. For example, basic science fields such as molecular biology and immunology have more citation relationships with clinical medicine, indicating that basic research provides biological and immunological support for aerobic exercise interventions. Applied disciplines such as health care and psychology pay more attention to the quality of life and mental health of patients. By citing clinical research results, the application of aerobic exercise intervention in rehabilitation treatment is further expanded.

Through the color depth time axis, we can observe the evolution path of different subjects' hot spots. Early research focused on molecular and biological mechanisms, and recently, the hotspots have gradually shifted to more applicable fields such as psychology and health care. This indicates that the focus of research has gradually shifted from exploring the basic

mechanism to optimizing the intervention effect and improving the quality of life of patients.

3.6. Keywords analysis

3.6.1. Keyword frequency and frequency analysis. Keyword frequency statistics chart and frequency statistics chart (Fig. 12B, D) show the most frequently occurring keywords in the selected literature, reflecting the main directions of current research in the field.

In the literature, the frequency of "knee osteoarthritis" was the highest, 116 times, accounting for 16.07%, which indicated that KOA was the core topic of current research and had high academic attention. This was followed by "aerobic exercise" 61 times (8.45%) and "hip" 58 times (8.03%). The high frequency of these keywords indicates that aerobic exercise and its effects on joint sites are important topics for research in this field. In addition, "pain" (54 times, 7.48%), "management" (48 times, 6.65%), "exercise" (48 times, 6.65%) also had a high frequency.

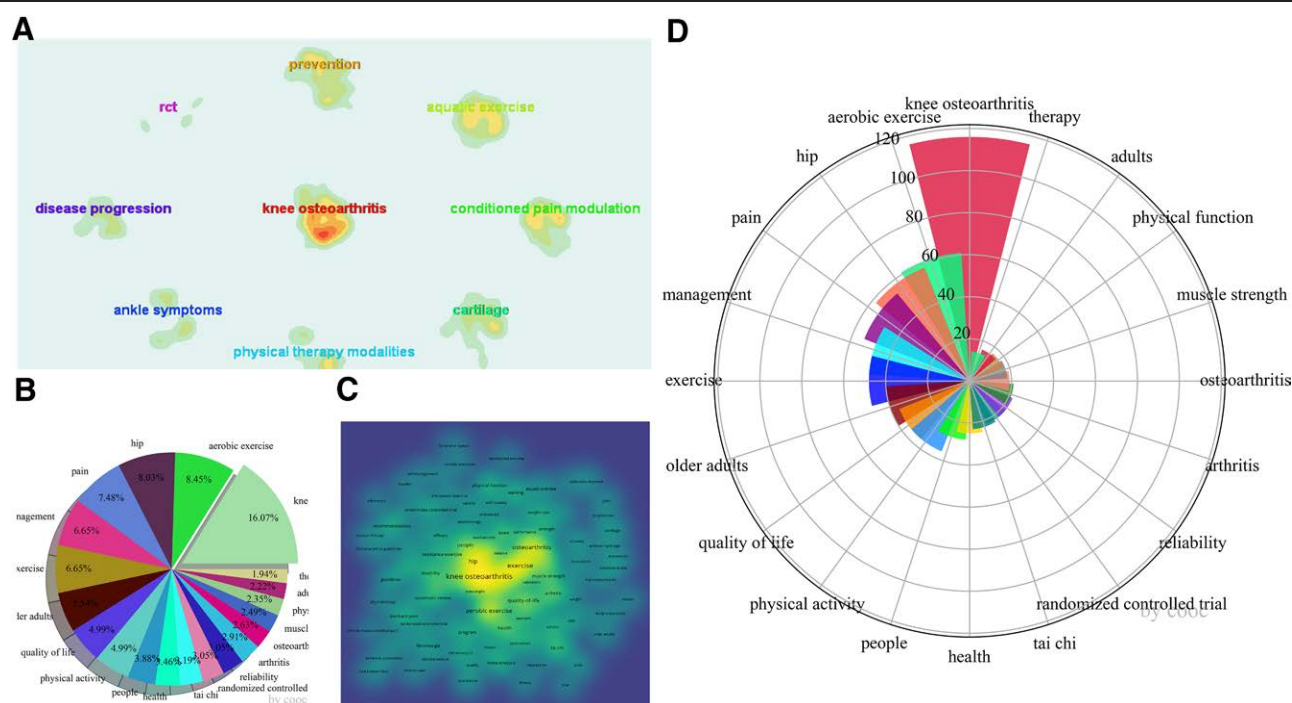


Figure 12. Keywords: synthesis diagram. (A) Displays a distribution of keywords. Terms including “rct,” “prevention,” “aquatic exercise,” “disease progression,” “knee osteoarthritis,” “conditioned pain modulation,” “ankle symptoms,” “physical therapy modalities,” and “cartilage” are presented in a spatial layout, with density variations indicating prevalence of research focus. (B) Presents a pie chart of keyword proportions. Categories like “hip,” “aerobic exercise,” “knee,” “pain,” “management,” “exercise,” “older adults,” “quality of life,” “physical activity,” “people,” “health,” “tai chi,” “randomized controlled,” “reliability,” “arthritis,” “osteoarthritis” occupy defined percentage shares, reflecting relative research attention. (C) Shows a heatmap-based keyword visualization. Keywords are displayed with color intensity denoting frequency or relevance, centered on terms such as “knee osteoarthritis,” illustrating the distribution pattern of research-associated keywords. (D) Depicts a radar chart of keywords. Terms including “knee osteoarthritis,” “therapy,” “adults,” “physical function,” “muscle strength,” “osteoarthritis,” “arthritis,” “reliability,” “randomized controlled trial,” “tai chi,” “health,” “people,” “physical activity,” “quality of life,” “older adults,” “exercise,” “management,” “pain,” “hip,” “aerobic exercise” are plotted. Radial axes and sector sizes represent research emphasis and interconnections, with sector angles and radii corresponding to keyword attributes and influence.

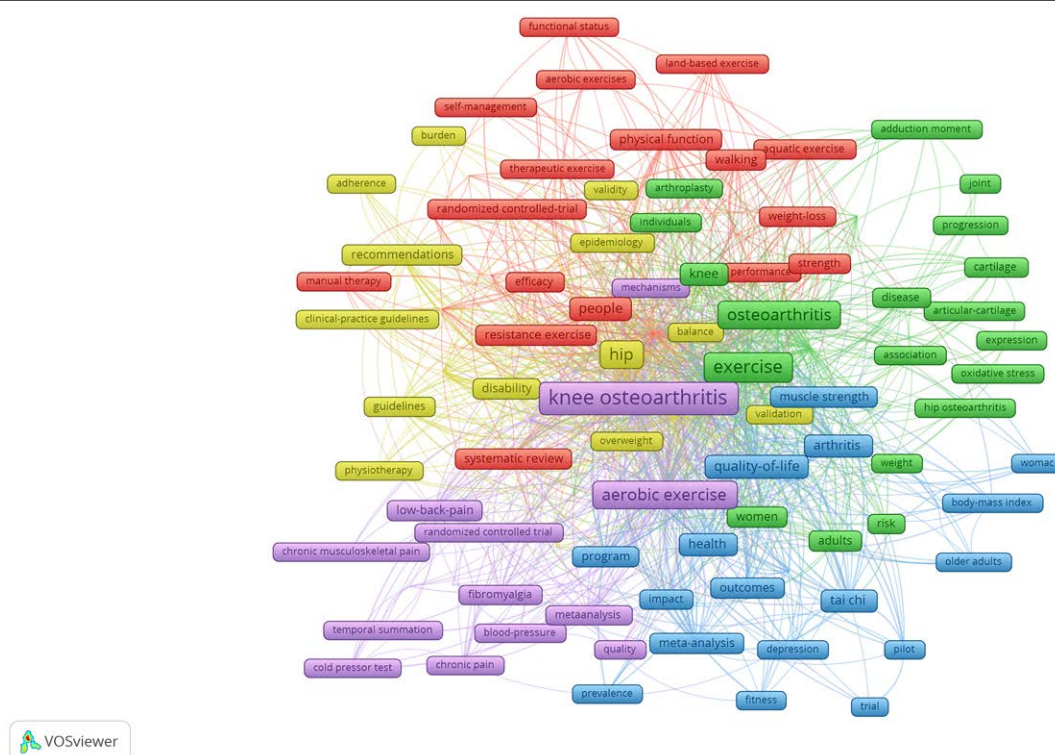


Figure 13. Keywords co-occurrence map.

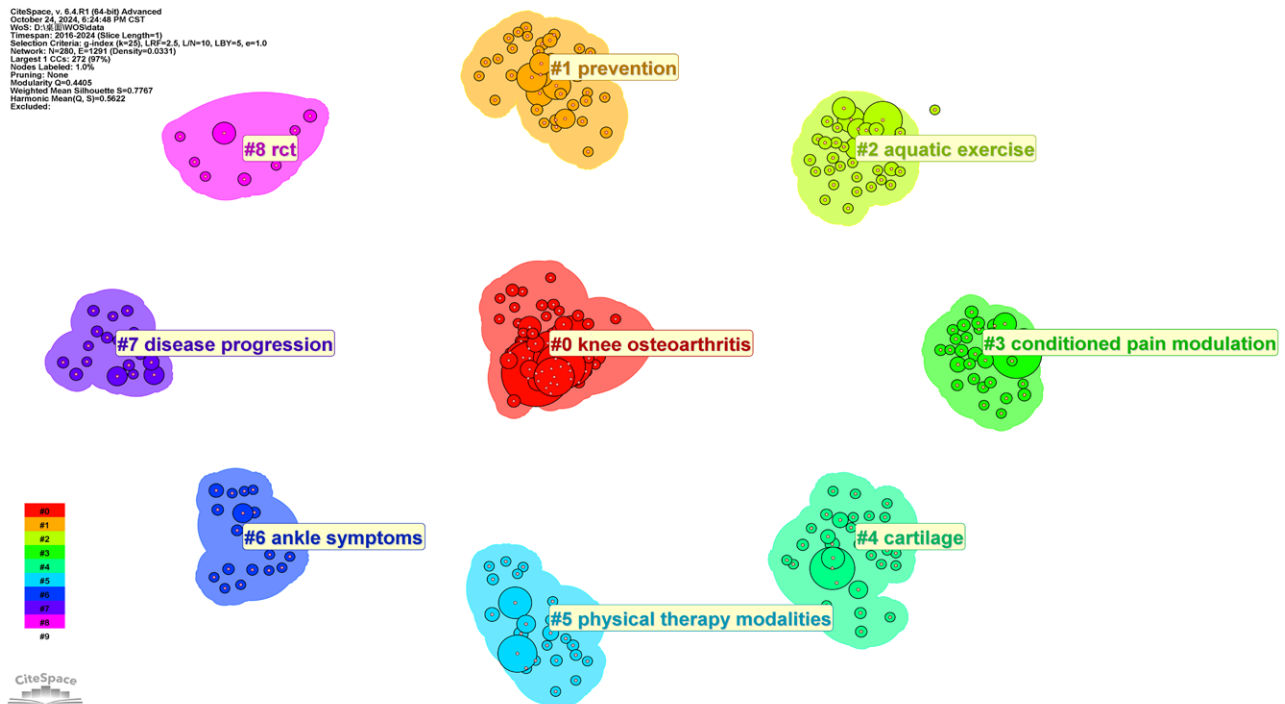


Figure 14. Keywords cluster map.

These words reflected the focus on pain management and exercise intervention in the literature. Some medium frequency keywords, such as “older adults” (40, 5.54%) and “quality of life” (36, 4.99%), indicated that part of the research focused on the elderly group and how to improve the quality of life through interventions.

This frequency analysis reveals that researchers are highly concerned about knee OA and its interventions, and also indicates that aerobic exercise and its effects on related parts are a key direction of research.

3.6.2. Keyword co-occurrence analysis. The keyword co-occurrence graph (Fig. 13) reveals the close correlation between different research topics by analyzing the co-occurrence relationship between keywords. In the co-occurrence graph, “knee osteoarthritis” was the most important core node, and there were a large number of connections between “knee osteoarthritis” and other keywords such as “exercise,” “pain” and “hip.” This shows that research on knee OA often involves exercise, pain management, and hip related content, reflecting the inherent connection between these topics.

Keywords such as “aerobic exercise,” “physical activity” and “quality of life” also showed a high co-occurrence frequency. This suggests that research should focus not only on specific conditions but also on the combined effects of exercise on improving patients’ quality of life. The co-occurrence analysis also showed the association between keywords such as “randomized controlled trial (RCT)” and “tai chi” and core nodes, indicating that RCT were mostly used to verify the effectiveness of traditional exercise methods on KOA. In general, the keyword co-occurrence analysis revealed the close relationship among multiple topics such as KOA, exercise intervention and pain management, emphasizing the importance of comprehensive intervention from multiple perspectives.

3.6.3. Keywords cluster analysis. Keyword cluster analysis (Fig. 14) further reveals the hot spots in the research field and the content of each cluster topic through the clustering of keywords in the literature. The value of cluster module

$Q = 0.6128$ indicates that the research structure of the whole field has a relatively clear community division. In general, $Q > 0.3$ indicates that there is an obvious community structure in the network, while $Q = 0.4405$ in this view indicates that the research topic of directional movement has a clear boundary, and different research directions form their own independent research groups. The cluster profile coefficient $S = 0.7767$ further revealed the internal closeness and discrimination of the clusters. $S > 0.7$ indicates that the clustering effect is very ideal and the theme consistency is high. This means that within each cluster, the co-occurrence relationship between keywords is very strong and the content of the research topic is highly concentrated.

We found that cluster #0 – “knee osteoarthritis” was the largest and core cluster in the literature, containing keywords such as “exercise,” “pain,” and “hip,” reflecting that the research on KOA mainly focused on the effect of exercise intervention on pain. And its effect on relevant joint sites. The research content of this cluster focused on exploring how exercise can improve the symptoms of knee OA, as well as the biomechanical and efficacy evaluation of exercise interventions. Cluster #1 – “prevention.” This cluster contained key words related to prevention, indicating that scholars were studying how to prevent the occurrence and development of KOA through effective intervention. This part of the research focuses on how to prevent the occurrence of knee joint diseases through early intervention and exercise. Cluster #2 – “aquatic exercise” focused on the intervention effect of aquatic exercise on knee OA. Water exercise has the characteristic of reducing the burden on the joint, so it has been widely studied in improving the pain and function of patients with KOA. This clustering highlights the positive role of water exercise in reducing joint stress and improving mobility. Cluster #3 – “conditioned pain modulation.” This cluster reflects another hot spot in the research, namely the mechanism of conditioned pain modulation, which mainly involves the neuromodulation mechanism of pain in patients with KOA. This suggests that researchers are focusing on the neurophysiological basis of pain management. Cluster #4 to

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 Timespan: 2016-2024 (Slice Length=1)
 Selection Criteria: g-index (k=25), LRF=2.5, LN=10, LBY=5, w=1.0
 Network: N=280, E=1291 (Density=0.53331)
 Largest CCs: 272 (97%)
 Nodes Labeled: 1.9%
 Pruning: None
 Modularity Q=0.4405
 Weighted Mean Silhouette S=0.7767
 Harmonic Mean(Q, S)=0.5622
 Excluded:

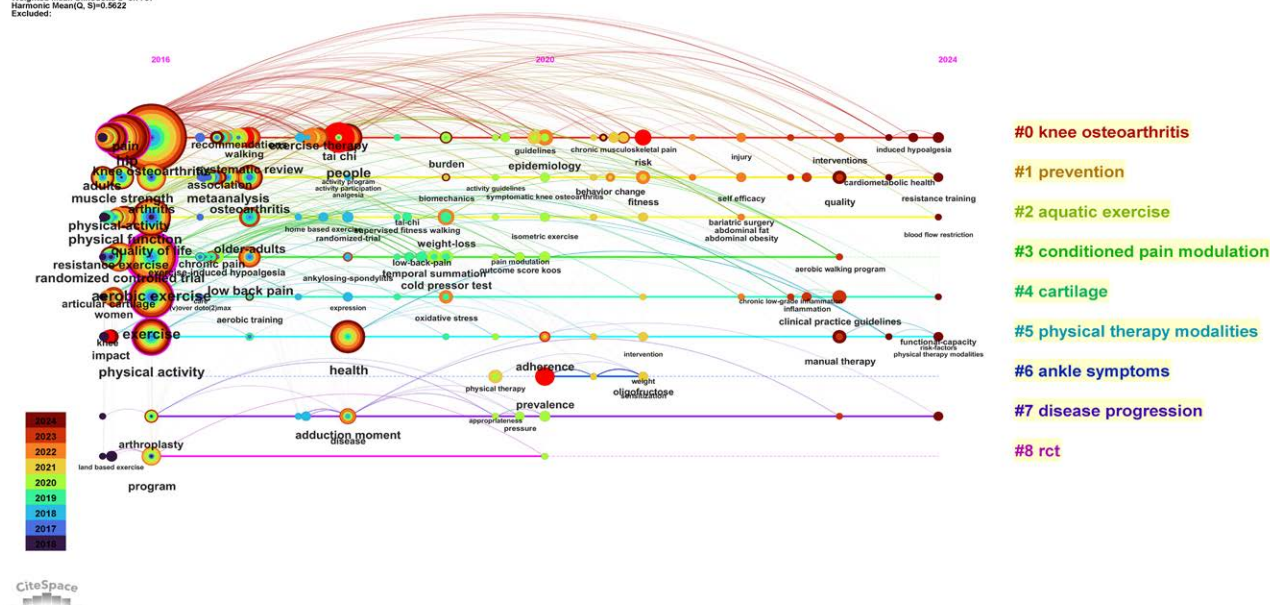


Figure 15. Keywords time chart.

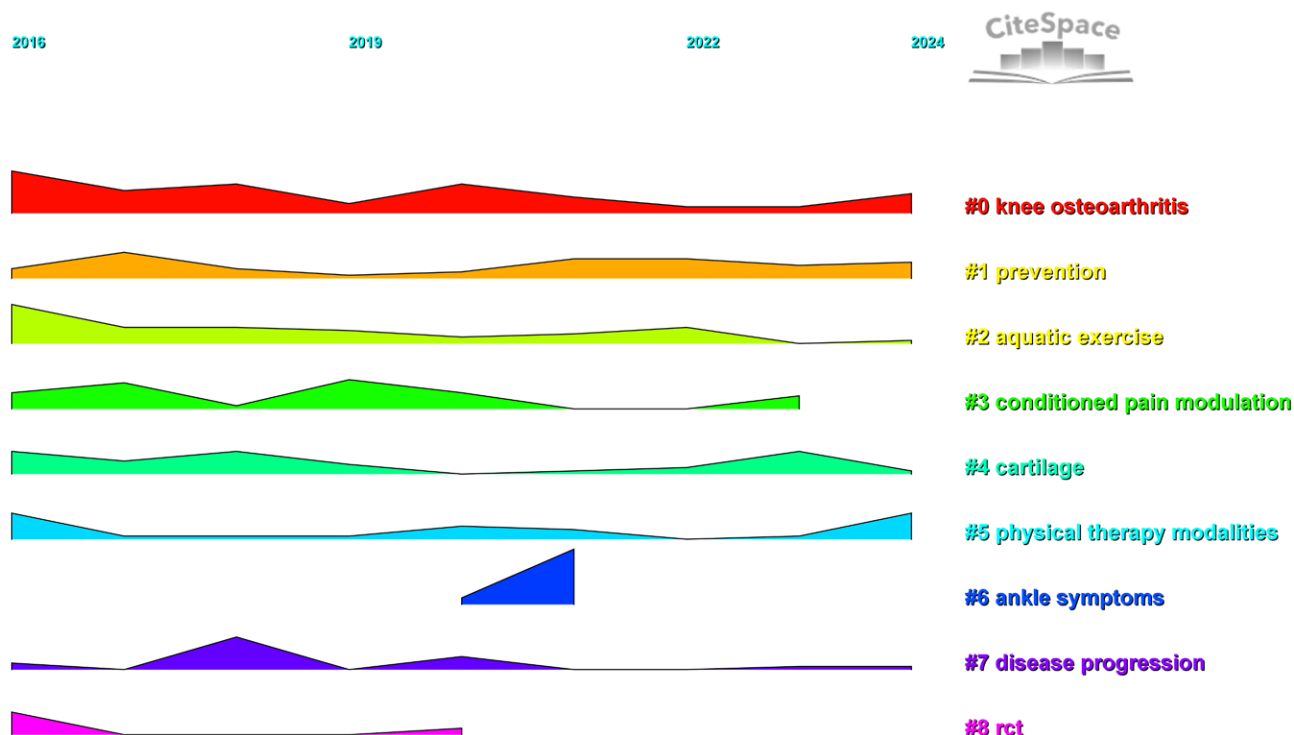


Figure 16. Keywords mountain map.

#8 involved “cartilage,” “physical therapy modalities,” “ankle symptoms,” “disease progression” and “RCT.”

These clusters discussed the cartilage degeneration in KOA, the effect of physical therapy on improving patients’ symptoms, the symptoms of ankle joint in the disease, and the progression mechanism of the disease. The keyword

cluster heat map shows the research heat of different clusters and their distribution in time. For example, keywords such as “knee osteoarthritis,” “prevention” and “aquatic exercise” showed high research popularity throughout the study, which reflects the continued attention of the academic community to these topics.

3.7. Hot spot evolution and trend analysis

3.7.1. Keywords timeline analysis. Figure 15 shows that KOA is the leading research hotspot throughout the whole period (2016–2024), and its frequency maintains a high-level on the time line, and has been at the top of research. The continuity and extensiveness of such studies indicate the critical role of knee OA in exercise intervention and health. “Prevention” and “Aquatic Exercise” had a stable research heat from 2016 to 2024, showing the importance of preventive measures and low-impact exercises (such as aquatic exercise) in the prevention and treatment of joint diseases. Conditioned pain modulation has received a slight increase in research heat in the later period, reflecting the increasing focus on understanding and controlling the pain mechanisms associated with KOA. The research heat of physical therapy modalities has been steadily increasing since 2020, indicating that with the development of physical therapy interventions, its effectiveness in symptom relief has attracted more and more attention. Disease progression began to show high research interest in 2017, and then the heat fluctuated less. This type of research focuses on observing the progression of knee disease in order to better understand and manage risk factors for the disease.

3.7.2. Hotspot clustering and trends. KOA, as the main research direction, formed a close co-occurrence network with a variety of keywords, such as “randomized controlled trial,” “pain regulation,” and “prevention.” These associations reflect the wide application of diverse research methods and interventions in this field. The cluster analysis mountain map (Fig. 16) provides the change of each research topic from 2016

to 2024, and the color blocks in the figure indicate the research intensity of each hotspot in different time periods. For example, the cluster of KOA remained significant throughout, reflecting its continued attention over time. As can be seen from the figure, the focus of research gradually shifted to diversification and fine differentiation over time. For example, from the general research on KOA and exercise intervention in the early stage to the gradual increase in the focus on specific interventions, such as water exercise and physical therapy means, this reflects the research need for more personalized and targeted treatment methods. With the evolution of research hotspots, the frequency of keywords such as “prevention” and “Physical Activity” also showed a certain trend of change, which indicated that the focus of research was not only limited to the treatment itself, but also gradually shifted to how to reduce the incidence of KOA through daily prevention.

These analyses show that research hotspots focus on a variety of interventions and preventive measures for KOA, and over time, research hotspots have gradually developed in a more in-depth and detailed direction, such as the effectiveness and specific mechanisms of different interventions. At the same time, these trend analyses also help to guide future research directions and resource allocation in this field.

3.7.3. Keyword attention and growth rate analysis. Figure 17A shows the change trend of attention of each key word related to knee arthritis in the academic literature from 2016 to 2024. It can be seen that the 2 keywords “Knee Osteoarthritis” and “Osteoarthritis” have always maintained a high degree of attention throughout the time range, among which “Knee

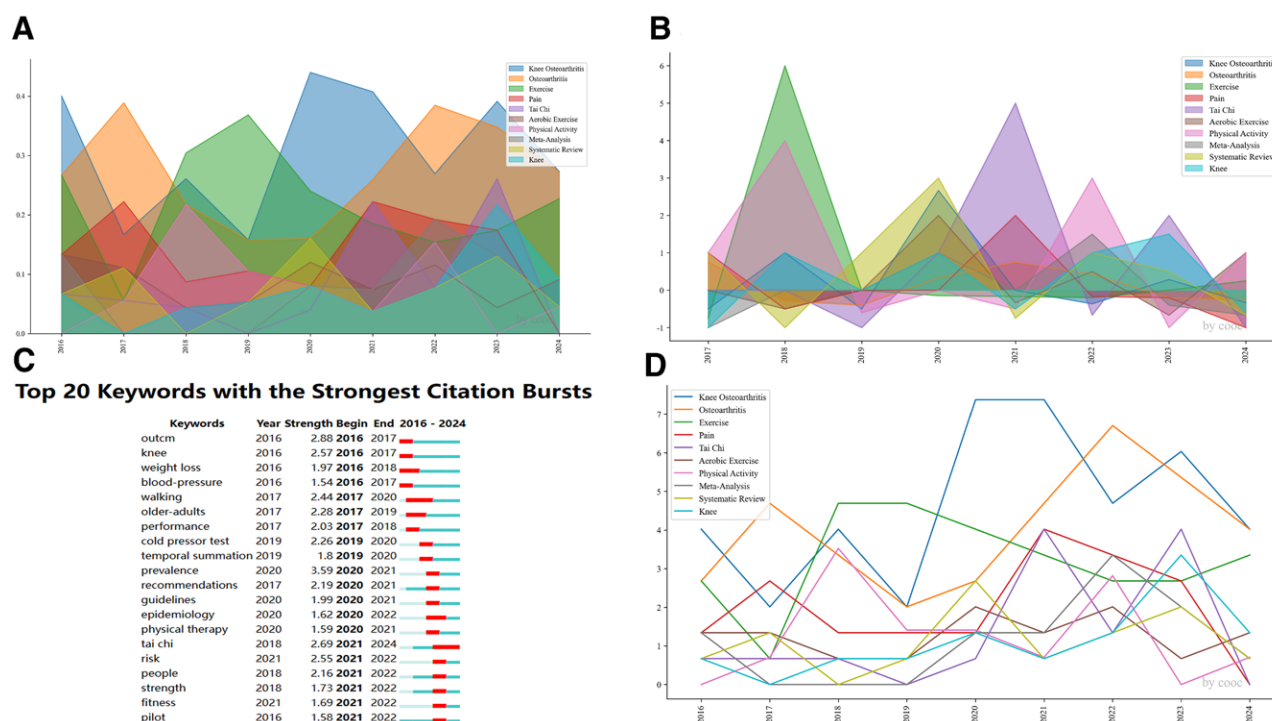


Figure 17. Keywords development trend comprehensive chart. (A) Presents the attention trend of knee - arthritis - related keywords in academic literature from 2016 to 2024. Different colors correspond to distinct keywords, with the right-side legend specifying color-to-keyword mappings. “Knee Osteoarthritis” and “Osteoarthritis” sustain close attention, with “Knee Osteoarthritis” prominent in 2021 and 2023. Keywords like “Exercise,” “Pain,” and “Tai Chi” show fluctuations; “Aerobic Exercise” becomes a 2022 focus, highlighting research on its knee arthritis effects. (B) Displays keyword growth-rate changes across years. In 2018, “Exercise,” “Tai Chi,” and “Aerobic Exercise” hit growth-rate peaks, reflecting surging research on non-pharmacologic, alternative knee-arthritis symptom relief. (C) Shows 2016 to 2024 keyword burst intensities. “Prevalence” (intensity 3.59) spikes in 2020, making the disease-prevalence study a hotspot. “Tai chi” (intensity 2.69, 2021–2024) gains attention as a rehab intervention. “Walking” and “weight loss” trend 2016–2020, emphasizing exercise’s role in knee OA symptom improvement. (D) Via time-weighted analysis, tracks research-hotspot evolution. “Knee osteoarthritis” and “osteoarthritis” draw continuous high attention, peaking around 2020, tied to aging-society-related research demand. “Tai chi” and “aerobic exercise” have emerged lately, reflecting focus on traditional exercise and aerobic exercise in middle-aged and elderly KOA. Post-2021 “meta-analysis” and “systematic review” fluctuations signal rising demand to summarize existing KOA exercise-intervention research. KOA = knee osteoarthritis.

Osteoarthritis” is particularly prominent in 2021 and 2023 and becomes a research hotspot. At the same time, the popularity of keywords such as “Exercise”, “Pain” and “Tai Chi” also fluctuated, especially in 2019 and 2022.

Different colors represent different keywords, and the legend on the right lists the specific keywords for the corresponding color. “Aerobic Exercise” became a significant focus in 2022, indicating a greater concentration of research on the effects of aerobic exercise on knee arthritis during the year.

Figure 17B shows the growth rate changes of each keyword in different years, providing more refined information on the evolution of research hotspots. It can be seen that in 2018, the growth rates of keywords such as “Exercise”, “Tai Chi” and “Aerobic Exercise” peaked, indicating that there was a significant increase in the amount of research on these topics during the year. This trend may be related to increased public interest in non-pharmacologic treatment options and increased interest in alternative therapies to relieve symptoms of knee arthritis.

3.7.4. Keyword outbreak and hot spot change detection.

3.7.4.1. Keywords outbreak analysis. In the keyword burst analysis, we identified keywords that suddenly appeared at high frequency during a specific time period, and these words reflected hot changes in academic research. The identification of keyword bursts helps to understand the development of academic interest in different topics within a field. Figure 17C shows the burst intensity of keywords and their duration from 2016 to 2024.

For example, “prevalence” appeared significantly in 2020, with an intensity of 3.59, reflecting that the study on disease prevalence had become a major research hotspot in this year. “tai chi” also had a high emergence intensity (2.69) between 2021 and 2024, indicating that Tai Chi received more attention as a rehabilitation intervention during this period. In addition, keywords related to exercise and lifestyle modification, such as “walking” and “weight loss,” had a high occurrence during 2016 to 2020, reflecting the importance of exercise intervention in improving the symptoms of knee OA.

3.7.4.2. Time-weighted hot spot change detection. Time-weighted hotspot change detection shows the dynamic evolution of research hotspots in time series. Through time weighting, we can observe the trend of research heat on different topics over time. As shown in Figure 17D, from 2016 to 2024, the research in the 2 core fields of “knee osteoarthritis” and “osteoarthritis” continued to attract high attention, and peaked around 2020. This shows that with the advent of the aging society, the research on osteoarthritis has always been a continuous hot spot in the academic community. In addition, keywords such as “tai chi” and “aerobic exercise” have gradually emerged in recent years, reflecting the increasing attention to traditional exercise intervention and the role of aerobic exercise in improving KOA in middle-aged and elderly people.

There were some significant fluctuations in “meta-analysis” and “systematic review” after 2021, indicating that the demand for summarizing and analyzing the existing research results has increased in recent years, especially in the systematic evaluation of the effects of different exercise interventions on KOA.

4. Discussion

For the first time, this study used bibliometrics and knowledge mapping analysis to systematically review the research on aerobic exercise intervention in KOA from 2014 to 2024 and revealed the research hotspots and trends in this field. Keyword analysis revealed that “pain relief” and “joint function improvement” are frequently discussed topics, indicating significant research interest, joint function improvement and quality of life improvement.

We found that countries such as the United States, China, and Australia dominated the research in the field of KOA aerobic exercise intervention, forming an international scientific research cooperation network with the United States as the core. The concentration of scientific research resources and extensive international cooperation in the United States have promoted the development of this field, especially in evidence-based medicine and systematic review. Especially in 2021, the intensity of its cooperation has increased significantly. This finding not only included established research powers (e.g., the United Kingdom and the Netherlands), but also emerging Asian research powers (e.g., China and South Korea). This finding reveals that research investment and bilateral cooperation on exercise intervention in developed countries compared with less developed countries may be caused by national economy. At the same time, James A. Evans (2014) also found a similar conclusion in his study, which found that developed countries had more health studies than less developed countries. Because medical research requires a lot of resources, it is largely related to national wealth.^[28]

China has also shown a significant growth trend in the research of KOA aerobic exercise intervention in recent years, especially playing an important role in applying traditional exercise forms, such as tai chi and Baduanjin, to KOA intervention. China’s node started in blue in 2019, turned green in 2020, and changed to red in 2021, indicating its gradual increase in the number of research collaborations and international participation in these years. This may be related to the Chinese government’s emphasis on scientific research in the health field and its gradually increasing investment in the field of KOA. We found that the National Natural Science Foundation of China has provided tremendous support and impetus to open access research in China by focusing on frontier disciplines and technological development, as well as Yusheng Li (2022) research, which has found that OA-related project applications have rapidly increased by 6 times in 2019 compared to 2010. This coincides with our knowledge mapping study, and Yusheng Li (2022) found that the number and size of researchers in the field of OA research have increased significantly since the establishment of the Department of Health Sciences of the National Natural Science Foundation of China, contributing to high-level research outcomes.^[29]

We also reveal the core authors and research team in the field of aerobic exercise intervention for knee OA, with representative authors including Bennell Kim L. And Brosseau Lucie et al The research networks led by these authors not only rank in the top in the number of papers published, but also have an important impact on the research direction and hotspots in this field. Bennell Kim L. ‘s research focuses on the clinical efficacy of aerobic exercise, with special attention to exercise intensity, joint function improvement, and patient pain relief, forming an interdisciplinary academic team. Brosseau Lucie focused on systematic evaluation and evidence-based medicine, focusing on data analysis of clinical trials and RCT, providing rigorous evidence-based support for KOA intervention. Multiple collaborative clusters were formed among these core authors, indicating the relatively concentrated distribution of academic resources among different research groups, and also showing the clustering effect of research in this field. In addition, authors such as Herzog Walter and Wang Chenchen are also influential in their own research fields, covering biomechanics, exercise protocol development, and personalized management.

In terms of institutions, the University of Melbourne and University of Southern Denmark have the largest number of publications in this field. The former has made significant contributions to the research on the combination of aerobic exercise and rehabilitation medicine, and the latter focuses on biomechanics and the evaluation of the effect of exercise intervention. Shanghai University of Traditional Chinese Medicine has a prominent performance in the intervention of KOA with traditional exercise forms such as Tai chi and Baduanjin, which

illustrates the gradually enhanced international status of traditional medicine in this field. Harvard University in the United States has great advantages in long-term intervention effects and evidence-based medicine, and its research team focuses on developing personalized exercise programs for KOA patients. On the whole, the main institutions have formed a number of transnational and cross-regional cooperation networks, among which the University of Melbourne has close cooperation with institutions such as the University of Southern Denmark and Aalborg University. The international cooperation in biomechanics, sports rehabilitation and clinical research has been gradually strengthened, which has promoted the internationalization of this field.

We also found some interesting results in terms of citation analysis. For example, the studies of McAlindon et al (2014) and Bannuru et al (2019) were widely cited, showing their important theoretical contributions and clinical guiding value in the field of KOA intervention. The study by McAlindon et al may have involved innovative therapeutic approaches and a new clinical research framework, thus showing significant citation strength between 2016 and 2019. The study by Bannuru et al showed a significant increase in citation intensity after 2021, possibly because it provided a new evidence-based basis for exercise intervention programs for KOA patients. Through citation co-occurrence analysis, it can be observed that these high-impact literatures formed a dense citation network in the research network of KOA aerobic exercise intervention, especially in the literatures involving RCT, pain management and functional improvement, showing a high co-citation frequency. Taken as a whole, these highly cited literatures built the theoretical and practical foundation of this field and became the core reference for subsequent research. From the citations, it can be seen that the research on aerobic exercise intervention for KOA is gradually developing in the direction of multidisciplinary integration. This trend reflects the synergy between different disciplines that provides a more comprehensive perspective for research, contributing to a deeper understanding and optimization of interventions. At the same time, it also points out the direction for future research to further improve the therapeutic effect of aerobic exercise intervention by integrating multidisciplinary knowledge. At the same time, research by Kentaro Onishi showed that multidisciplinary and multimodal treatment strategies combined with early detection and prevention strategies provide the best benefits for patients.^[30]

Keyword analysis revealed the topic distribution and hot spot evolution in KOA aerobic exercise intervention research. The keyword frequency analysis showed that “knee osteoarthritis” and “aerobic exercise” were the core themes, reflecting the focus of research in this field on improving the quality of life and functional performance of KOA patients through exercise. Other high frequency keywords such as “pain,” “quality of life” and “randomized controlled trial” indicated that research focused on pain relief, function improvement and intervention effectiveness verification. Co-occurrence analysis also revealed a strong association among the key words. The high co-occurrence frequency of “tai chi” with “quality of life” and “pain” indicated that the research on traditional exercise forms in improving KOA symptoms has received extensive attention. In addition, keyword cluster analysis revealed several major clusters of research topics, including “biomechanical effects,” “exercise intensity and frequency,” and “psychological management of patients.” In recent years, the emergence of keywords such as “personalized exercise intervention,” “long-term efficacy” and “neuropathic pain modulation” indicates that the research direction has gradually expanded from basic efficacy evaluation to multidimensional and more complex application research. In addition, we further conducted a keyword time-weighted analysis and found a future upward trend of “Exercise.”

This study provides valuable insights into the research progress and trends of aerobic exercise interventions in the treatment

of KOA, but there are some limitations. First, it relies solely on the Web of Science database and may have missed relevant studies from other sources. Only publications in English were included, which may have led to language bias. Third, the study focused on bibliometric indicators without assessing the methodological rigor or clinical data of the included studies, limiting the practical significance of the findings. Heterogeneity in methodology and intervention protocols across studies may affect the robustness of the conclusions. Future studies should address these issues and provide more comprehensive and applicable results.

5. Conclusions

In this study, we systematically sorted out the research progress and hot trend of aerobic exercise intervention in patients with KOA. Current studies have shown that aerobic exercise intervention has a significant effect on KOA patients in terms of reducing joint pain, improving joint function and improving quality of life, especially traditional exercise forms of low to moderate intensity (such as tai chi and water exercise) are prominent in the intervention effect. Through bibliometrics and knowledge mapping analysis, it was found that the research hotspots gradually evolved from the efficacy evaluation of a single form of exercise to the attention of personalized intervention, long-term efficacy and pain management mechanism, showing that future research is positively tilted to individualized and dynamic exercise prescription design.

From the perspective of international cooperation and interdisciplinary trends, the United States, China, Australia, and other countries dominate the research on KOA aerobic exercise intervention, form a transnational collaboration network, and promote the integration of exercise physiology, rehabilitation medicine and clinical trials. This trend provides a scientific support for exploring the intervention mechanism and optimizing the intervention strategy of KOA. However, different intervention methods, inconsistent evaluation indicators, and lack of long-term intervention studies still affect the reliability and applicability of the intervention effects. Future research should strive to standardize intervention design and develop personalized exercise intervention programs combined with the dynamic changes of patients, so as to achieve long-term effects.

Overall, this study reveals the broad application potential of aerobic exercise intervention for KOA patients, provides a reference for clinical practice and future research, provides theoretical support for optimizing exercise intervention strategies for KOA, and points out the future development direction in multidisciplinary collaboration and dynamic intervention models.

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